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Leakiness and flow capture ratio of insect pectinate antennae

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The assumption that insect pectinate antennae, which are multi-scale organs spanning over four orders of magnitude in size among their different elements, are efficient at capturing sexual pheromones is commonly made but rarely thoroughly tested. Leakiness, i.e. the proportion of air that flows within the antenna and not around it, is a key parameter which depends on both the macro- and the microstructure of the antenna as well as on the flow velocity. The effectiveness of a structure to capture flow and hence molecules is a trade-off between promoting large leakiness in order to have a large portion of the flow going through it and a large effective surface area to capture as much from the flow as possible, therefore leading to reduced leakiness. The aim of this work is to measure leakiness in 3D-printed structures representing the higher order structure of an antenna, i.e. the flagellum and the rami, with varying densities of rami and under different flow conditions. The male antennae of the moth *Samia cynthia* (Lepidoptera: Saturniidae) were used as templates. Particle image velocimetry in water and oil using 3D-printed scaled-up surrogates enabled us to measure leakiness over a wide range of equivalent air velocities, from 0.01 m s^{-1} to 5 m s^{-1} , corresponding to those experienced by the moth. We observed the presence of a separated vortex ring behind our surrogate structures at some velocities. Variations in the densities of rami enabled us to explore the role of the effective surface area, which we assume to permit equivalent changes in the number of sensilla that host the chemical sensors. Leakiness increased with flow velocity in a sigmoidal fashion and decreased with rami density. The flow capture ratio, i.e. the leakiness multiplied by the effective surface area divided by the total surface area, embodies the above trade-off. For each velocity, a specific structure leads to a maximum flow capture ratio. There is thus not a single pectinate architecture which is optimal at all flow velocities. By contrast, the natural design seems to be robustly functioning for the velocity range likely to be encountered in nature.

1. Introduction

In many moth species, olfaction is a key sense for successful reproduction. After emergence, female imagoes call for males by releasing tiny amounts of sex pheromones [1] that are carried by the wind over large distances of up to several hundred metres [2] and detected by males with the help of their antennae [3]. When encountering an antenna of a male moth, the pheromone molecules must first be captured, before being carried to and detected by the sensors located within the sensilla of the antenna. In the Saturniidae family, these antennae have a particular shape and are termed pectinate antennae (figure 1a). They are composed of one main branch, the flagellum, which supports many secondary branches, the rami. Numerous small hairs, the sensilla, hosts of the sensors, are located on the rami. In our work, we focused on the antenna of one single species, *Samia cynthia*.

In this work, we study the pheromone capture step, when molecules in the air reach the surroundings and the surface of an antenna. This step consists of a mass transfer between the air and the surface of the antenna. At low concentrations, which is the case for moth sex pheromones, mass transfer depends only on the

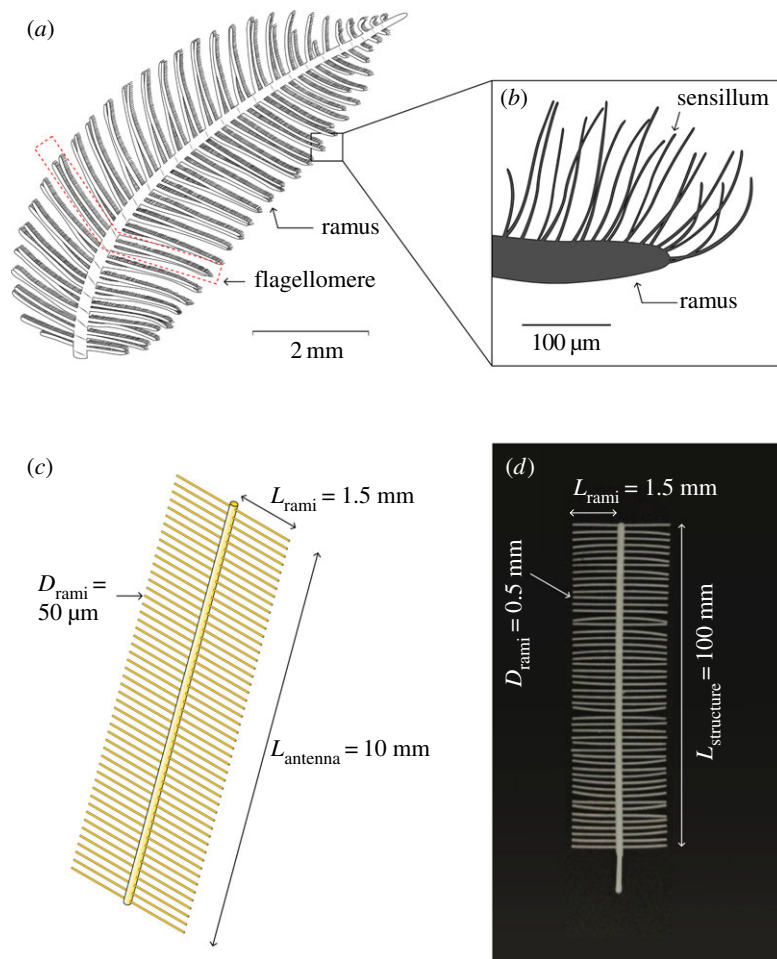


Figure 1. Illustration of an antenna of *Samia cynthia* and the 3D-printed surrogate structure. In (a), the structure composed of the flagellum and rami is visible. Note that the rami are not completely parallel and overlap at their tips. (b) Focus on the tip of a ramus to reveal the sensilla. (c) Simplified model of the gross morphology of the natural antenna. (d) The actual printed surrogate structure (named Str₅₀). Like all of our 3D-printed structures, this structure is scaled-up by a factor of $\alpha = 10$ compared with the biological model. This model bears $N_{\text{rami}} = 2 \times 50$ rami and has an effective surface area of $S_{\text{eff}} = 7.5 \text{ mm}^2$.

flow of air through the antenna and on the diffusive coefficient of the pheromone in air. At very high concentrations, mass conservation and, thus, the respective densities of air and pheromone would need to be taken into account. Given the complex shape and multi-scale nature of the antennae, determining the flow of air around them through analytical models is a very difficult task. One of the difficulties lies in the fact that an antenna is a complex object with finite dimensions. Thus, air can flow between the rami and sensilla, or around the entire antenna. We are particularly interested in the proportion of air that flows within the antenna instead of around it, called leakiness in accordance with several previous studies [4,5]. In other contexts, porosity, which is closely linked with leakiness (at a given velocity, a very porous structure allows more fluid to pass through it than a less porous one), has been shown to play an important role in the fluid dynamics around permeable discs [6]. It is an important measure in the context of olfaction, as it gives an upper limit of the proportion of pheromone molecules that can be captured and detected by an organ. In this study, we limit our work to the leakiness of the antenna and how it is affected by the number of rami. Leakiness measures how much pheromone goes through the antenna and is, thus, susceptible to be captured. It gives an upper limit of what is available to the sensilla. However, to determine the mass capture by the antenna, the diffusive process at the sensillum scale is also very important as well as how the pheromone molecules get

into the pores of the sensilla, once they have been captured by the antenna.

Leakiness is a complicated property to determine because it depends both on the macroscale and on the microscale of the antennae. Furthermore, the leakiness of a permeable structure varies with air velocity. Cheer & Koehl [7] determined the leakiness of an antenna, but they reduced the number of rami to two. This trick was actually a powerful one as it allowed them to calculate the interaction between the two rami while keeping the geometry simple enough to be able to solve it analytically. Indeed, an analytical approach is likely to be too complex in the case of an entire structure. Consequently, the leakiness of the entire organ has been experimentally measured, but without any variation in the geometry of the antenna [5,8–10].

As we were interested in the influence of the entire architecture of antennae, we decided to measure the leakiness with particle image velocimetry (PIV) on scaled-up, 3D-printed antennae. PIV is mainly used in experimental fluid mechanics and in biology to obtain time-resolved velocity measurements and related properties in fluids [11,12]. We adapted the fluid viscosity and velocity to keep the Reynolds number in correspondence with the situation in nature, which ensured that the dynamics for real antennae and for our artificial structures are similar [13,14].

A moth antenna is a multi-scale object: its length is approximately 1 cm whereas its sensilla are only a few micrometres

thick. Thus, there are four orders of magnitude between these two extremes. We resorted to 3D printing to produce antennae with various geometrical parameters. 3D printing is a very convenient tool as it gives greater design freedom than traditional manufacturing processes. It also allows single customized objects to be produced. Currently, no additive manufacturing technology is able to span over four orders of magnitude [15]. As a consequence, we limit our modelling to one level, the structure composed of the flagellum and rami (figure 1a), and we choose as a first approximation to neglect the effect of the second, lower level on the flow around the antennae. This second level is the microstructure (figure 1b), which is composed of one ramus and its sensilla. In other words, we neglect the effect of the sensilla on the flow. This assumption will be discussed. We changed the number of rami to investigate whether the natural configuration of 50 rami of 50 μm diameter is in some way optimal in terms of leakiness and flow capture for varying wind speeds. Changing the number of rami results in a change in effective surface area (or frontal area) of the antenna. We postulate that an increase in the effective surface area allows an antenna to support more sensilla, which is where the chemical detectors are located.

2. Material and methods

2.1. Real and surrogate antennae

2.1.1. Morphology of the moth antennae

We first proceed with the characterization of the main morphological features of *Samia cynthia* antennae. Cocoons were ordered from the Office for Insects and their Environment, Guyancourt, France. After one to two weeks, the imago emerged and were directly put in a freezer. The moths were then dried at room temperature for a month. We used electron microscopy (Quanta 450; FEI, Hillsboro, OR, USA) to obtain pictures of the antennae and the software Inkscape (v. 0.91, <http://www.inkscape.org>) to measure the length of the antennae. The measurements were done on five antennae taken from five male moths.

We found that an antenna is 1 cm ($9.25 \text{ mm} \pm 0.4 \text{ mm}$ (mean $\pm$ s.e.)) long on average (figure 1a). The flagellum is divided into smaller segments called flagellomeres and each flagellomere bears two pairs of rami, one on each side, perpendicular to the flow direction. The most distal flagellomeres may bear only two rami, one on each side. So, on average, one antenna has 100 rami perpendicular to the flow, 50 (49.2 ± 2.3) on each side. To measure the length of the rami, we used 37 rami. We neglected the shortest ones at the tip of the antennae (eight or nine rami depending on the exact number of rami on each antenna) and the four closest ones to the base of the antennae because they were surrounded by the glue used to fix the antennae onto the support. It was thus not possible to obtain a good measurement of their length. We found that a ramus is approximately $1.35 \text{ mm} \pm 0.3 \text{ mm}$. As we measured the length of the rami on a two-dimensional image and because the rami are slightly curved in the direction perpendicular to the plane of the picture, we underestimated the length of the rami. We thus considered them equal to 1.5 mm in our artificial models.

2.1.2. 3D-printed antennae

We modelled the antenna as a row of cylinders (the rami), linked by a perpendicular one (the flagellum). The cylinders mimicking the rami were all straight and all had the same length in order to simplify the geometry (figure 1c). We also considered that the cylinders had the same diameter all along their length, even though there is a slight narrowing close to the tip of the natural

rami. However, the exact shape of the tip of a cylinder has been shown to have a small influence on the fluid dynamics [16,17].

Because of 3D-print limitations, but also to observe the velocity field in more detail, we scaled-up our artificial structures by a factor of 10 relative to the natural ones and adjusted both the fluid velocities and viscosities. The Navier–Stokes equations ensure that configurations of similar geometries with equal Reynolds numbers have similar flow profiles: for example, their non-dimensionalized velocities or accelerations are equal [18]. This peculiarity of the Navier–Stokes equations is convenient as it allows the geometries to be scaled-up by adjusting the two other parameters, which are the fluid velocity and the fluid viscosity (equation (2.1)). To change the fluid viscosity over the entire range of values of biological relevance, we had to change the nature of the fluid (i.e. viscosity). To cover the entire range of interesting Re values, we worked in both water and rapeseed oil. We then adjusted the fluid velocities to keep the Reynolds number constant,

$$Re = \frac{Lv}{\nu}, \quad (2.1)$$

where Re is the Reynolds number (non-dimensional), L is the characteristic length (m), v is the characteristic velocity (m s^{-1}) and ν is the kinematic viscosity of the fluid ($\text{m}^2 \text{s}^{-1}$).

From now on, the 3D-printed artificial antennae surrogates will be referred to as *structures* and the different designed *structures* will be described by the term Str_x (Str standing for 3D-printed surrogate structure). For instance, Str_{50} was designed to mimic the real antennae, scaled by a factor 10, as it also had 50 rami of diameter 500 μm on each side of the flagellum (thus, 100 rami in total). Furthermore, we wanted to investigate the influence of the size of the gap between the rami and the influence of the effective surface area S_{eff} so we built structures with {30, 40, 60, 70} rami on each side of the flagellum, respectively labelled $\{\text{Str}_{30}, \text{Str}_{40}, \text{Str}_{60}, \text{Str}_{70}\}$ with the rami evenly spaced along the flagellum. These complementary structures cover geometries beyond the range of biological designs and were specifically chosen to explore a wide range of geometries. We defined the *effective surface area* as the projected surface area of the structure of rami (neglecting the projected surface area of the flagellum, which is kept constant),

$$S_{\text{eff}} = 2 \cdot N_{\text{rami}} \cdot D_{\text{rami}} \cdot L_{\text{rami}}, \quad (2.2)$$

where N_{rami} is the number of rami on one side of the flagellum, D_{rami} is the diameter of the rami and L_{rami} is the length of the rami.

Furthermore, we defined the total surface of the structure S_t , bounded by its lateral extent, as

$$S_t = 2 \cdot L_{\text{struc}} \cdot L_{\text{rami}}, \quad (2.3)$$

where L_{struc} is the length of the structure.

Given the specific shape of each of our structures, we resorted to 3D printing to build them. Indeed, 3D printing is convenient to build customized objects [15]. However, there are still some limitations depending on the specific type of 3D printing. We utilized poly-jetting with an Objet Eden 260 (Stratasys, Eden Prairie, MN, USA), which is a material-jetting 3D-printer: droplets of material are jetted to build the object layer by layer. The benefit of the Eden 260 is the use of a support material which allows designs with overhanging parts. However, the support material must be removed and this step can be tricky with numerous thin cylinders, as is the case here. The voxel resolution of this 3D printer is around $\approx 42 \mu\text{m}$ diameter times a thickness of $\approx 16 \mu\text{m}$, giving the smallest printed details of around a few hundreds of micrometres [19]. All structures were designed using Openscad (v. 2015.03-2) [20]. Using a single row of rami on each side of the surrogate allowed us to conveniently take off the support material.

We chose the VeroClear (Stratasys, Eden Prairie, MN, USA) material to build our artificial structures. Thus, the surface properties of our structures are different from those of a real moth antenna,

Table 1. Summary of the PIV experiments and corresponding velocities in air, water and oil.

velocity in <i>air</i> (m s^{-1})	0.01	0.02	0.05	0.1	0.2	0.5	1	2	5
velocity in <i>water</i> (mm s^{-1})					1.28	3.21	6.41	12.8	32.1
velocity in <i>oil</i> (mm s^{-1})	3.21	6.41	16	32.1	64.1	160			

which is composed of a mix of chitin and wax substances. However, at our scales (tens of micrometres in air for the moth antenna and hundreds of micrometres in water for the artificial structures), it has been shown that the no-slip boundary condition is valid [21,22] and that the roughness of the surface has a negligible effect at our range of Reynolds numbers [23]. We are thus confident that the surface properties of the antenna or the artificial structure did not affect the fluid dynamics. We also assume no fluid–structure interactions, which makes our experiments independent of the mechanical properties of the materials.

2.2. Flow measurements

We used PIV [24] to visualize the velocity field. This experimental technique consists of seeding a fluid with micrometric particles that are assumed to follow the fluid motion. A laser sheet is then used to illuminate the particles in a given plane. A camera in the perpendicular direction records the position of the particles at regular intervals. The fluid velocity field is extracted from these pictures through the use of a correlation algorithm which compares one picture with that taken at the next time step [25].

2.2.1. Scaling up and description of the experiments

We investigated a wide range of relative velocities between the air and the antenna to cover air flows through an antenna on a resting moth or, equivalently, flows through an antenna during moth flight. Thus, we used biologically relevant air velocities of 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2 and 5 m s^{-1} . Because of the technological limitations of the poly-jetting process, we scaled-up our artificial structure by a factor 10 as previously explained. Equation (2.1) shows that, in order to keep the Reynolds number constant, the velocity and/or the viscosity of the fluid must be adjusted. Adjusting velocities in our PIV experiments is bounded by practical considerations related to achievable accelerations and the required minimum time for measurements. Therefore, instead of working in air (kinematic viscosity of $15.6 \text{ mm}^2 \text{ s}^{-1}$ at 25°C and 10^5 Pa) we decided to run our experiments in water and in oil for their appropriate kinematic viscosities ($1 \text{ mm}^2 \text{ s}^{-1}$ for water at 20°C and $50 \text{ mm}^2 \text{ s}^{-1}$ for heated rapeseed oil at 29°C). This method is commonly encountered in fluid dynamics and flight aerodynamics [13].

The water and oil velocities (table 1) were adapted in order to keep the Reynolds number equal to those in air according to

$$v_{\text{water}} = \frac{1}{\alpha} \frac{v_{\text{water}}}{v_{\text{air}}} v_{\text{air}} \quad (2.4)$$

and

$$v_{\text{oil}} = \frac{1}{\alpha} \frac{v_{\text{oil}}}{v_{\text{air}}} v_{\text{air}}, \quad (2.5)$$

where v_{water} is the far-field water velocity, v_{water} is the water kinematic viscosity, v_{oil} is the far-field oil velocity, v_{oil} is the water kinematic viscosity and $\alpha = 10$ is the scaling factor.

We ran two experiments for each structure. In both cases, the flow was parallel to the z -axis, the laser sheet was parallel to the xz -plane and the camera was in the y -direction (figure 2). We positioned our structures perpendicular to the flow.

For experiments in water, the equivalent velocities to the lowest velocities in air are too small to obtain any good

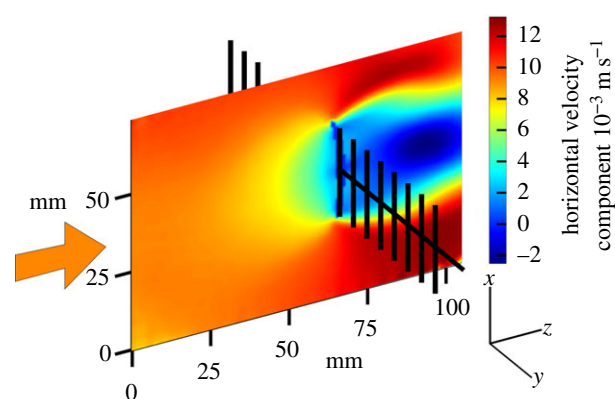


Figure 2. Experimental set-up and definition of variables. This velocity field was obtained for the structure Str_{50} at a velocity of 3.21 mm s^{-1} (equivalent air velocity: 0.5 m s^{-1}) in water. The arrow shows the direction of the flow.

measurements. Likewise, at large velocities in air, the velocities in oil are too high to be practical. We thus switched between rapeseed oil and water to be able to explore the entire velocity range. Note that, at the equivalent velocities in air of 0.2 and 0.5 m s^{-1} , the measurements in water and oil overlap, allowing us to observe the consistency of our measurements.

2.2.2. Particle image velocimetry set-up

The structures were immersed into a tank filled with water or rapeseed oil (figure 3). To generate a constant flow, we fixed the structure on a support and moved the tank of fluid at the desired velocity. In this configuration, the laser source and the camera were also fixed. Motion of the tank was realized by means of a motorized linear axis (LST1000D-T3; Zaber Technologies, Vancouver, British Columbia, Canada). We added a cover made of hard plastic at the surface of the water to prevent the formation of waves during the acceleration phase. We then obtained a tank of water or oil moving at the desired velocity. Hollow glass particles covered with silver ($10 \mu\text{m}$; SUGS-10; Dantec Dynamics, Skovlunde, Denmark) were used to track the fluid movements. Our denser structure Str_{70} had a centre-to-centre inter-rami distance of 1.4 mm . As the rami had a diameter of 0.5 mm , the edge-to-edge inter-rami distance was then equal to 0.9 mm , which is 100 times larger than the particle size. Thus, we can conclude that the particles are small enough for our experiment.

A 1.5 mm wide laser sheet was generated by a continuous laser source (MGL-F; 532 nm ; 2WCNI, Changchun, China) equipped with a cylindrical lens. Details of the velocities used in the two experiments in water and oil are summed up in table 1. The tank was 50 cm long and 30 cm wide and the artificial structures were 3 cm wide. Thus, at our range of Reynolds numbers, we could neglect the wall effects on the flow [26,27].

A high-speed camera (Phantom 9.1; Vision Research, Perth, Australia) was used to record the motion of the particles. We wanted to have a complete view of the flow and equipped the camera with a 35 mm lens (AF-S DX; Nikon, Tokyo, Japan) of aperture $1/4$. The field of view was then equal to $105 \times 77 \text{ mm}$. The resolution of the pictures was set at 1632×1200 pixels. One hundred pictures per second ($\Delta t = 10 \text{ ms}$) were recorded except at 32.1 , 64.1 and 160 mm s^{-1} . At these last three velocities,

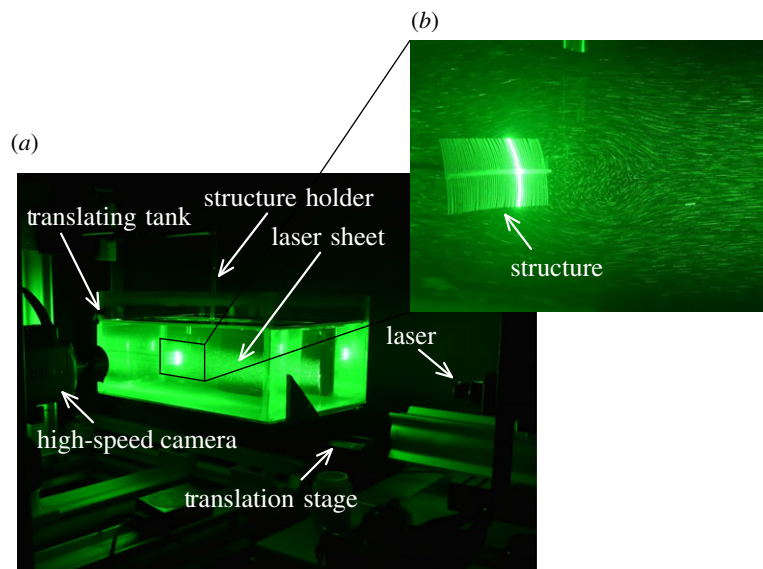


Figure 3. Set-up of the PIV experiment. (a) Set-up of the PIV experiment, (b) laser sheet. We can see that the laser sheet is wide enough to cover at least a complete inter-rami space.

we reduced the exposure time to 5 ms to prevent particles appearing as line segments rather than points.

Raw image processing was done using the software package Davis (Davis 8.3; Lavision, Gottingen, Germany). Processing consisted of extracting the velocity field from the pictures taken by the camera. The first step was subtraction of the background noise resulting mainly from the reflection of the laser light by the static object in the flow. It allowed us to keep only the flowing particles. We then applied a mask at the location of the structure and we chose an adaptive cross-correlation algorithm with 50% overlapping. The interrogation windows were chosen between 24 and 32 pixels. The capacity of the cross-correlation to accurately determine the velocities depends on the displacement, in number of pixels, of the particles between two correlated pictures. The maximum measured velocity, $v_{\max}$, is defined according to equation (2.6) and the minimum one, $v_{\min}$, is defined according to equation (2.7). The ratio of minimum and maximum velocities indicates the span of the dynamical range of velocity measurements,

$$v_{\max} = \frac{L}{\Delta t} \frac{L_{IW}}{N_L} \quad (2.6)$$

and

$$v_{\min} = \frac{L}{\Delta t} \frac{0.1}{N_L}, \quad (2.7)$$

where L is the length of the field of view ($L = 105$ mm), N_L is the number of pixels ($N_L = 1632$ px), L_{IW} is the size, in pixels, of the interrogation window ($L_{IW} = 24$ – 32 px) and Δt is the time between two pictures.

The term 0.1 (in pixels) in equation (2.7) refers to the minimum pixel displacement that the algorithm can detect: 0.1 means it can detect sub-pixel displacement. In the case of $\Delta t = 10$ ms, the minimum velocity is 0.613 mm s^{-1} , which is almost equal to the minimum velocity in water (0.641 mm s^{-1}) and higher than the minimum velocity in oil (0.21 mm s^{-1}). To achieve a larger velocity measurement dynamical range, we decided to decrease the minimal detectable velocity by increasing the time between two pictures, as explained above. This was done by taking one picture every 50 pictures and one every 70 pictures in the case of the lowest oil velocity, giving a minimum detectable velocity of, respectively, 0.0123 mm s^{-1} and 0.0088 mm s^{-1} . The maximum velocity for $\Delta t = 10$ ms is equal to 147 mm s^{-1} and is sufficient for our entire range of velocities. For a given velocity, on a given structure, we determined the time evolution of the velocity field, using several subset pairs of images and a suitable time gap to obtain an

appropriate dynamical range of measurable velocities. As we studied steady-state flows, we then averaged the time-resolved velocity fields to determine the steady-state velocity field.

2.2.3. Estimation of the leakiness and definition of the flow capture ratio

We consider our structure to be a homogeneous permeable object. This means that the leakiness is also homogeneous and can be assessed from the evaluation of flow velocities in a single plan. However, because the structure has a finite size, edge effects occur, influencing the leakiness locally. As it has a high aspect ratio, i.e. it is longer than it is wide, edge effects are relatively more important in the width direction, that is, in the direction along the rami (x -axis, figure 2). Thus, we can expect more variations along a ramus (x -axis) than along the flagellum (y -axis). We thus decided to measure the flow in an xz -plane located approximately in the middle of the structure. There are, however, important velocity variations according to whether a ramus is included or not in this plane. In our experimental set-up, the laser sheet has a given width of 1.5–2 mm. Indeed, it is difficult to produce very thin laser sheets. An ideal infinitely thin laser sheet would also encounter only a few particles and, thus, could not produce good enough pictures. Owing to the width of the laser sheet, our measurements provided a mean of the velocity fields of many planes, including planes with and without a ramus (figure 3b); this gave us a mean velocity field in a middle section of the structure.

Our method to measure leakiness, Le , consists of looking at the spatial decrease in the flux, ϕ , as we get closer to the structure. However, given the geometry of our measurement set-up, the exact flux at the precise position of the structure is inaccessible, because it is hidden by the structure itself. We thus integrated the measured flow through a segment parallel to the x -direction and as wide as the structure and followed the evolution of this flux as the segment was gradually moved closer to the structure (figure 4a). Because of the continuity of the flux, we could then determine the flux, ϕ_{rami} , within the structure (figure 4b). Then, we multiplied this two-dimensional flux by the length of the structure to get the total flux. We use this method in the rest of this work to determine the leakiness of the structure. Several estimations of the leakiness for the same structures showed that this method gives an estimate with an interval of leakiness of 5%. For each velocity, we did one trial and obtained one estimate of leakiness.

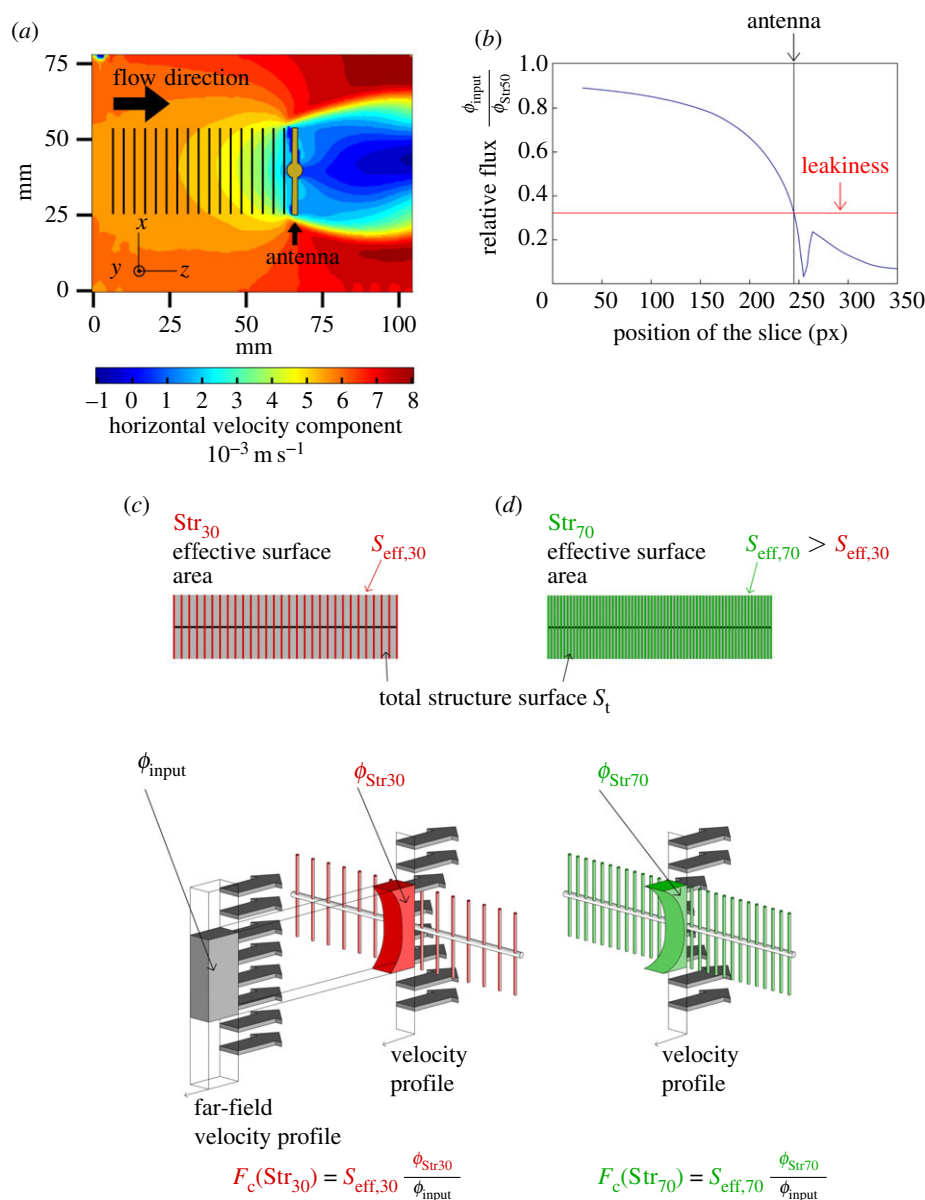


Figure 4. (a) Illustration of the slice method used to determine the leakiness of the structures. (b) Relative proportion of the flux through a section of a structure located upstream and downstream of it. The location of the structure is shown by the black line. The value of the leakiness, as determined by the relative flux at the entrance of the structure, is shown by the horizontal red line. This graph was drawn for the case of structure Str₅₀ at a water velocity of 6.41 mm s^{-1} (equivalent air velocity: 1 m s^{-1}). The measured leakiness is equal to 0.32. (c,d) Effect of the change of effective surface area on the flux inside the structure and estimation of the flow capture ratio.

In order to assess the effectiveness of a structure, we needed to take two related effects into account. Obviously, structures with large leakiness, Le , have a large flux passing through them, a prerequisite for pheromone uptake. At the same time, if the large leakiness is due to a very sparse structure, i.e. if there are few rami in the structure and the distance between the rami is large, the chances of capturing any of the pheromone molecules in the structure are rather low. Hence, we also needed to take into account the relatively available rami surface area. Taking both effects into account, from the combined estimation of the leakiness and of the structure effective surface area (S_{eff} ; see table 2 for details), we define a new parameter which reflects the efficiency of the artificial structure, the flow capture ratio, F_c . This ratio is defined as the product of two ratios: (i) the leakiness, Le , and (ii) the capture surface area ratio of a structure. This is the fraction of the effective surface area over the total surface area bounded by the lateral extent of the structure, i.e. S_{eff}/S_t (see table 2 for numerical values),

$$F_c = \frac{S_{\text{eff}}}{S_t} Le. \quad (2.8)$$

Table 2. The structures and their various geometrical parameters. The parameters displayed in *italics* are the ones equivalent to the moth antenna.

	N_{rami}	D_{rami} (μm)	S_{eff} (mm^2)	capture surface area ratio $\frac{S_{\text{eff}}}{S_t}$	scale
Str ₃₀	30	500	450	0.15	10
Str ₄₀	40	500	600	0.20	10
Str ₅₀	50	500	750	0.25	10
Str ₆₀	60	500	900	0.30	10
Str ₇₀	70	500	1050	0.35	10

Figure 4c,d illustrates the expected effect of a change of structure effective surface area, S_{eff} , on this flow capture ratio. For the structure Str₃₀ with 30 rami, having a relatively small effective surface area, $S_{\text{eff},30}$ (figure 4c), the leakiness is relatively important; this means

that the flux, $\phi_{\text{Str}30}$, is relatively large. We can, therefore, expect on the one hand a large flow capture ratio, F_{c30} , owing to the large leakiness of this sparse structure, but on the other hand we can also expect a small flow capture ratio, owing to the small effective surface area. We now consider a second structure, Str_{70} , with 70 rami and thus a larger effective surface area, $S_{\text{eff},70} > S_{\text{eff},30}$, as illustrated in figure 4*d*. The leakiness of this second structure is reduced compared with the first one, as the flux, $\phi_{\text{Str}70}$, between the rami decreases. This may translate into a reduction of the expected flow capture ratio, F_{c70} . At the same time, there might be an increase in the effective surface area owing to an increase in the flow capture ratio, F_{c70} . The trade-off between these two conflicting effects (i.e. an increase in effective surface area and a decrease in leakiness) greatly affects the flow capture ratio. Moreover, because of boundary layer effects, leakiness and, therefore, the flow capture ratio also greatly depend on the flow velocity.

3. Results

3.1. Particle image velocimetry measurements

When the flow encounters a structure, part of the fluid flows through the structure and part of the flow goes around the structure (figure 5). As expected, the wake of the structure is particularly affected by the change of flow regime, i.e. by the increase of flow velocity. At low velocities, the flow is laminar and symmetrical, and converges behind the structure (figure 5*a*). At high velocities, the wake is turbulent and we note the presence of shedding vortices, even though some vortices are still maintained behind the flagellum (figure 5*c*). In between, at medium velocities, a separated vortex ring is present in the wake of the structures (figure 5*b*) and the distance between the vortex ring and the structure depends on the fluid velocity. In their work on dandelion seed flight [6], the authors identified its porosity as a key design feature that enables the formation of the separated vortex ring. Our pectinate antenna is also a porous object and is another striking example of this new class of fluid behaviour around fluid-immersed bodies. Whether and how this might be beneficial in the context of olfaction needs dedicated work.

We have explored how the velocity fields in a cross section of the structures of constant ramus diameter are affected by the change of effective surface area. To do so, we used several structures with increasing numbers of rami with constant diameter (see table 2 for more details). The whole matrix of measurements is large, since we explore nine different flow velocities (from 0.01 m s^{-1} to 5 m s^{-1}) on five different designs, from Str_{30} ($N_{\text{rami}} = 30$ and $D_{\text{rami}} = 500 \mu\text{m}$, corresponding to an effective surface area, S_{eff} , of 450 mm^2) to Str_{70} ($N_{\text{rami}} = 70$ and $D_{\text{rami}} = 500 \mu\text{m}$, corresponding to an effective surface area, S_{eff} , of 1050 mm^2). Figure 6*a,b* illustrates how the change in effective surface area affects the flow around the structure at low velocity. Figure 6*c,d* illustrates how the change in effective surface area affects the flow around it at high velocity. We limit the examples to the extreme cases. We use these measurements to estimate the evolution of leakiness with an increase in effective area, by integrating the evolution of the flux as we get closer to the structure. We determined that structures with numerous cylinders, i.e. more than the natural design (Str_{60} and Str_{70} ; respectively, 60 and 70 cylinders), had a lower leakiness than those with fewer cylinders than the natural design (Str_{30} and Str_{40} ; respectively, 30 and 40 cylinders) at a given air velocity, as expected (figure 7). It is worth noticing that the scaling using constant Reynolds numbers works reasonably well: in the region

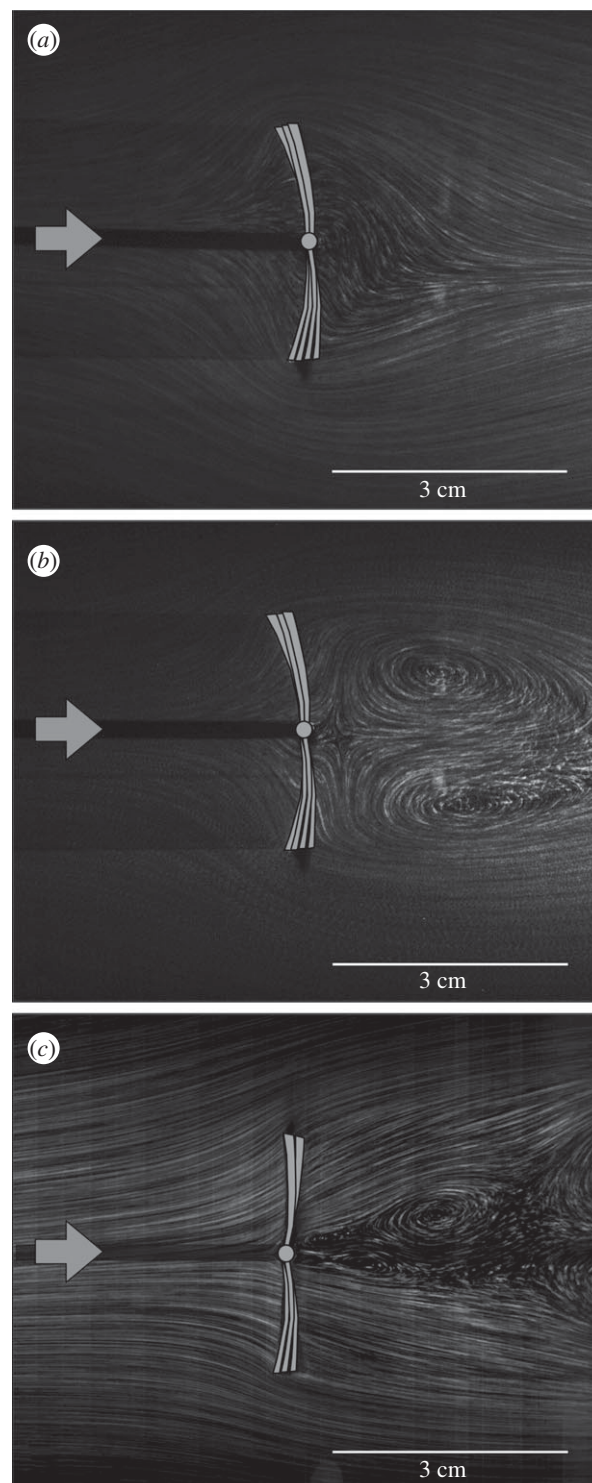


Figure 5. Flow around the surrogate structures, seen from the side. (a) Laminar flow around structure Str_{40} at an oil velocity of 3.21 mm s^{-1} , equivalent to an air velocity of 0.01 m s^{-1} . (b) Stable separated vortex ring in the wake of structure Str_{40} at an oil velocity of 64.1 mm s^{-1} , equivalent to an air velocity of 0.2 m s^{-1} . (c) Unsteady wake of structure Str_{40} at a water velocity of 12.8 mm s^{-1} , equivalent to an air velocity of 2 m s^{-1} . The grey arrows show the direction of the flow.

where the ranges of Reynolds numbers in oil and water overlap, the water and oil data give almost the same leakiness (figure 7).

3.2. Fitting leakiness relationships

The observed trends (figure 7) and the fact that the leakiness is bounded by 0 and 1 suggest a general behaviour the

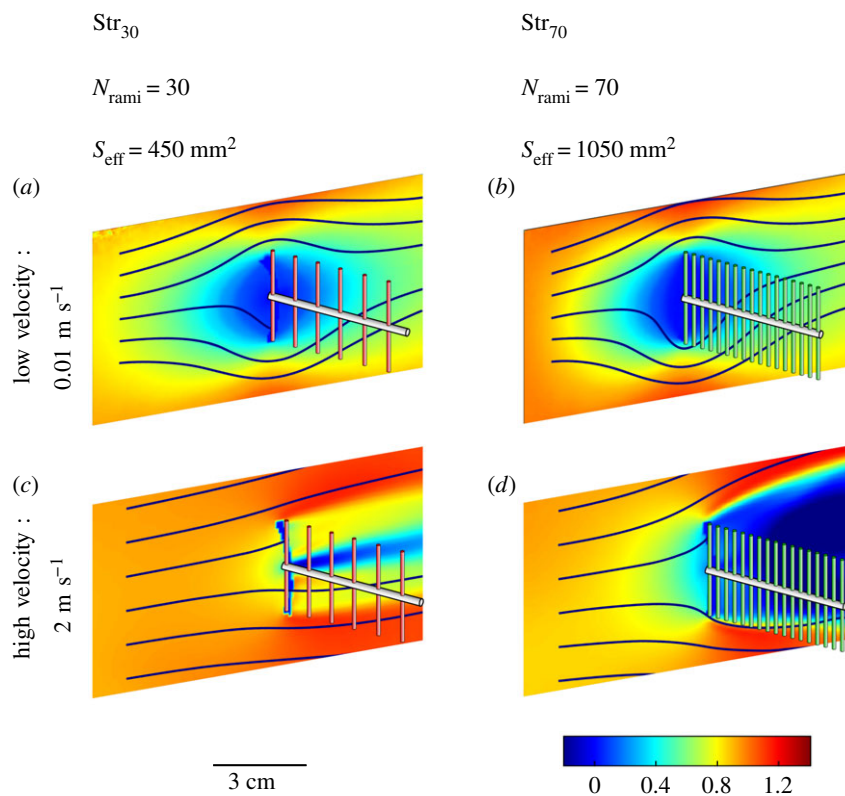


Figure 6. Influence of the effective surface area, S_{eff} , on the flow in a cross section of the 3D-printed scaled-up structures, with a constant diameter $D_{\text{rami}} = 500 \mu\text{m}$. Velocity fields in the case of: (a) low velocity and low effective surface area, (b) low velocity and high effective surface area, (c) high velocity and low effective surface area, (d) high velocity and high effective surface area. The velocities are normalized by the far-field corresponding velocity v_{∞} , which implies that the normalized velocities v/v_{∞} are non-dimensionalized velocities and vary mostly between 0 and 1.

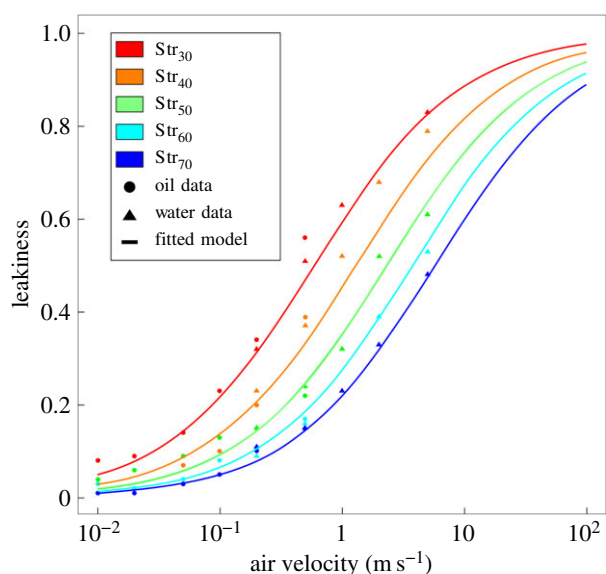


Figure 7. Leakiness of the structures and fitted curves. Each circle or triangle is an individual experiment.

increase of leakiness with velocity resembles a sigmoidal shape [8] and a reasonable fit can be obtained with

$$\text{Le} = \frac{v_n^\alpha}{1 + v_n^\alpha}, \quad (3.1)$$

where v_n is the normalized far-field velocity, i.e. $v_n = v_{\infty}/v_{50\%}$, and where $v_{50\%}$ is a normalizing velocity coinciding with the velocity at which leakiness is 0.5. This is an important point for the sigmoid as it is the location of the maximum of variation. Indeed, at very low and very

Table 3. Parameters obtained from the nonlinear regression for each structure. R^2 is the coefficient of determination.

structure	α	$v_{50\%}$	R^2
Str ₃₀	0.748	0.475	0.996
Str ₄₀	0.859	0.913	0.996
Str ₅₀	0.660	2.435	0.980
Str ₆₀	0.731	4.219	0.992
Str ₇₀	0.703	5.554	0.998

high velocities, the sigmoid is always close to 0 and 1, respectively. α can be used to tailor the slope to optimally fit the measured curves. We used the software RStudio (v. 1.0.136) to fit a sigmoid to each curve of leakiness using a least-squares method, the nonlinear regression function nls (table 3).

The $v_{50\%}$ values as a function of S_{eff}/S_t show a nearly exponential relation. A reasonable predictive function is

$$v_{50\%} = C \cdot \left(\frac{S_{\text{eff}}}{S_t} \right)^\beta. \quad (3.2)$$

Here, again, we used the nonlinear regression nls to determine the parameters and obtained

$$C = 97.43 \quad (3.3)$$

and

$$\beta = 2.69, \quad (3.4)$$

with $R^2 = 0.980$, the coefficient of determination.

We then carried out nonlinear regression on all the data to determine a common α for all the structures. The velocities $v_{50\%}$ were calculated according to equation (3.2) using the parameters in (3.3) and (3.4). We obtained

$$\alpha = 0.727, \quad (3.5)$$

with $R^2 = 0.983$, the coefficient of determination.

4. Discussion

4.1. Validity of the assumptions

A major postulate of our work, stated in the Introduction, is that increasing the effective surface area increases the number of sensilla and hence increases the capture probability of pheromone molecules. This postulate has a major implication. While sensilla require attachment points and hence a surface, increasing the effective surface area while keeping the total surface area constant is bound to fail: sensilla also need the empty spaces between the rami to function. Thus, while it is likely that the effective surface area is a determining variable, it might not be the only one defining the overall capture efficiency.

The capture of odours is a mass transport problem where both advection and diffusion play a role. However, in some cases, studying only one of the processes may be sufficient to assess the olfactory efficiency of a structure. For example, understanding the change in leakiness regime (underlying a change in the advection process of the molecules) of the olfactory appendages of underwater crustaceans is sufficient to understand how they capture odours [28,29]. Because the diffusion coefficient of molecules in water is low ($\approx 10^{-9} \text{ m}^2 \text{ s}^{-1}$ compared with $10^{-6} \text{ m}^2 \text{ s}^{-1}$ in air), crustaceans' appendages need closely packed structures. However, water cannot flow through them anymore, unless the appendages are actively moved to generate a flux. In a more general way, Waldrop *et al.* [30] showed that leakiness and advection are important to understand underwater olfactory appendages, whereas diffusion is required to understand olfactory antennae in air. They also showed that pectinate antennae are an intermediary case in which both phenomena should be studied. This phenomenon is supported by the estimation of the Péclet number, which is the ratio of the convective and diffusive terms and is equal to $Pe = uL/D$, with u the characteristic velocity, L the characteristic length and D the diffusive coefficient. If we use the diameter of the sensilla as the characteristic length, the Péclet number is 1000 in the case of the lobster, which lives in water [28], and 1 in the case of our moth. In the latter case, both convection and diffusion are thus important phenomena.

In our study, we focused on leakiness even though we know that diffusion is necessary to entirely understand the pectinate antennae. This will be investigated in a subsequent publication, once a study of the lower level with the sensilla and their sensors is carried out with equivalent depth. Moreover, given their size (micrometric sensilla on a millimetric ramus), the study of the lower level is challenging [31] but progress along those lines has been made very recently [32].

We measured the leakiness of the main structure of the antenna, i.e. of the antenna without its sensilla. This simplification is justified by the fact that the flagellum and the rami are expected to have the main influence on the air flow as already assumed in previous studies [7,33], especially at high velocities when the boundary layers become thin. However, the sensilla

may have a non-negligible impact and should also be considered when assessing leakiness. Such work has unfortunately not been done yet, given the complex multi-scale geometry of the antenna. Changing the diameter of the rami to include the effect of the sensilla might be a solution. We think that the trade-off that we showed between leakiness and the flow capture ratio still holds if the effect of the sensilla is taken into account.

We assumed the antennae to be perpendicular to the flow as this is generally assumed to be the case [7,8,10,33]. In fact, we do not know the orientation of Saturniidae moth antennae during flight but it has been observed that they are held perpendicular to the flow during sniffing [34]. It is thus natural to assume that antennae are oriented perpendicular or close to during flight. This approach simplifies the problem but still allows one to unravel the underlying physical mechanisms [6]. The orientation of the antennae in the flow is, however, an important parameter which most likely has a bio-physical relevance to the results. In some species of moths, their cylindrical-shaped antennae are used for flight control and are not perpendicular to the flow but have a specific orientation [35]. However, very recently, Spencer *et al.* [32] showed that antennal ramus orientation has little influence on small particle capture. The drag on cylinders placed in rows has also been studied from the sole point of view of fluid dynamics [36,37]. The cylinders were either two or three in a row and the orientation of the cylinders with respect to the flow varied. The authors of the two studies found that, for spacing of cylinders similar to that of the rami, the drag does not change by more than 20% between orientations of 0° to 50° with the vertical axis. Even though the Reynolds numbers were not the same as that in the case of the moth, these few published works suggest that our results should also have a reasonable degree of robustness regarding the orientation of the antenna.

Our model was also designed for a constant air flow. However, a periodic component of the flow appears during flight owing to the wing beats [38], which may be reinforced by the oscillation of the antennae [35]. Active sensing, generated by wing beats, also adds a periodic component. However, this periodic component does not change the airflow by an order of magnitude compared with the constant component alone [5]. Thus, we can assume that our work is still valid in the flows with a periodic component, self-induced or external, and is pertinent to understanding the behaviour of pectinate antennae in real air flows.

4.2. Similarity of the dynamics in different fluids

The scaling of the antenna and the shift from air to water and oil depend on the fact that keeping the Reynolds number identical to the natural situation is sufficient to ensure similar dynamics. The ranges of the water and oil velocities were defined so that their respective ranges of equivalent air velocities would overlap. The water and oil curves indeed fit broadly; in particular, the ranking between treatments, which is quite well conserved (figure 7).

One of the reasons for the observed discrepancies in the two fluids might be that the lowest velocities in water were too low to enable good measurement of the fluid dynamics. At these water velocities (0.641 mm s^{-1} and 1.28 mm s^{-1} , corresponding to air velocities of 10 cm s^{-1} and 20 cm s^{-1} , respectively), any small disturbance can have a big impact on the PIV measurement. For example, the laser sheet heats the structure and

generates upward water flow. This phenomenon is indeed especially noticeable at low velocities for two reasons: firstly, the displacement of the fluid structure is low and the heat generated by the laser sheet is thus dissipated in a small volume of water, giving rise to considerable temperature gradients. Secondly, the upward water velocity due to the difference in density is not negligible compared with the horizontal velocity.

Regarding the measurements made in oil, difficulties appeared at an equivalent air velocity of 50 cm s^{-1} . Here, the oil velocity is 160 mm s^{-1} . At this velocity, we had to decrease the shutter speed of the camera in order to have good images of the particles, rather than small lines due to their displacement. As a drawback, less light enters the camera and the images of the particles at the sides of the laser sheet are not sharp, leading to a loss of accuracy in the measurement of the velocity field.

4.3. Boundary layer superposition and its influence on leakiness

In all our measurements, leakiness increases with fluid velocity (figure 7). This result was expected because the boundary layer around an object tends to be smaller at high velocities, allowing more fluid to pass through the structure. We observed that the transition between an almost non-permeable structure at low velocities and a highly permeable structure at high velocities occurs at air velocities ranging from 10 cm s^{-1} to 1 m s^{-1} (figure 7). These air velocities are encountered by a moth in nature [39,40]. Thus, the geometry of the structures seems to be adapted to have a high leakiness at the air velocities at which the antennae are known to function.

We have shown that the structures with numerous rami had a lower leakiness than those with fewer rami at a given air velocity (figure 7). The variation in leakiness might be due to the extent with which boundary layers overlap. In the case of numerous cylinders, the cylinders are closer to each other and, therefore, more susceptible to having overlapping boundary layers than distant ones in which the boundary layers are more likely to have a weaker interaction with each other [41]. As a consequence, fewer pheromone molecules are available per sensillum in a denser structure. Sparser structures allow more air to pass through but the total surface area of the antenna is reduced. From the point of view of molecular diffusion, this also means that air flows at higher velocities, giving little time for the pheromone molecules to diffuse to the sensilla. However, without quantifying the diffusive process (which we plan to do in a subsequent publication), it is difficult to have more definitive statements.

4.4. Flow capture ratio

From the previous section, we understand the importance of the overlap of boundary layers around rami, eventually resulting in the reduction of leakiness of the structure when its effective surface area increases. The measurement of the leakiness alone, however, gives only a partial insight into the antenna efficiency. An antenna with a larger effective surface area is supposed to bear more sensilla in contact with the flow, and this might also influence the flow capture ratio defined previously. We now determine the overall flow capture ratio incorporating this effect. We simply define it as the product of the leakiness, Le , and the ratio of the effective surface area of the structure, S_{eff} , over its total surface area, S_t . This represents the ratio of a given far-field flux that can be caught by a

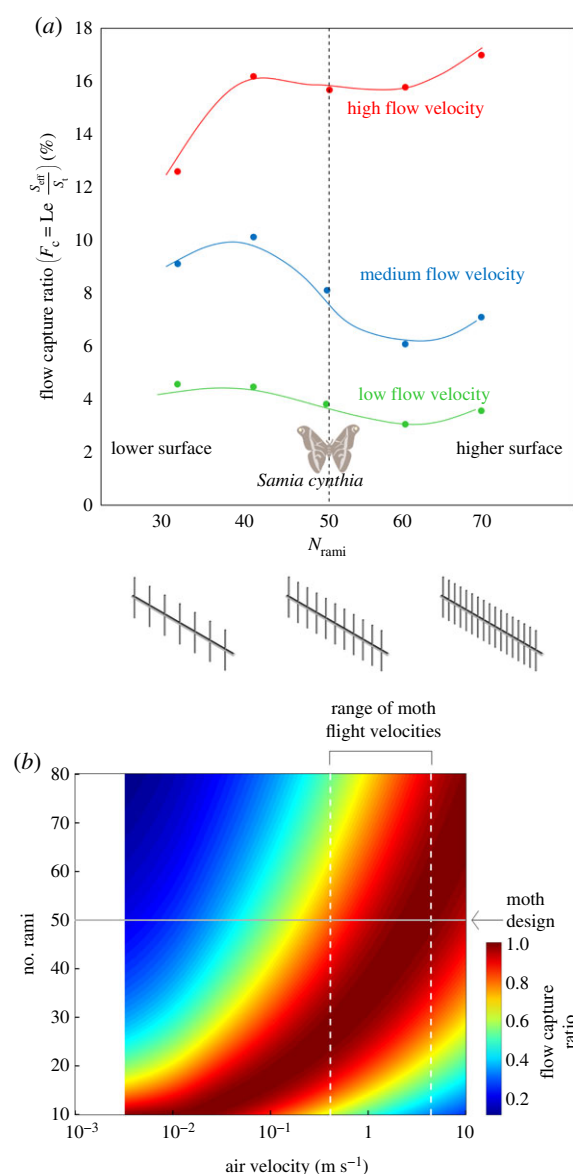


Figure 8. Variation of the flow capture ratio depending on the number of rami and the air velocity. (a) Variation of the flow capture ratio of the structures with changes of effective surface area at different flow velocities. (b) Normalized flow capture ratio depending on air velocity and number of rami. The normalization was done by dividing the flow capture ratio by the maximum flow capture ratio obtained for all the designs at the same air velocity. The white dotted lines show the range of moth flight velocities and the grey horizontal line highlights the performance of the natural design of the moth.

given surface area of rami-bearing sensilla. The calculation of this flow capture ratio highlights the trade-off between the two effects previously identified.

Figure 8a represents the dynamics of the flow capture rate for different effective surface areas. At low flow velocity, it appears that a smaller effective surface area is sufficient while a larger effective surface area seems to be more efficient at high velocity. We observe that the average effective surface area of the *Samia cynthia* antennae gives them an average efficiency over a wide range of fluid velocities.

From the sigmoid curves, we determined the leakiness of structures with different numbers of rami as well as their flow capture ratio. We used these fitted curves to calculate the leakiness of structures with a number of rami ranging from 10 to 80 and their respective flow capture ratios.

In order to compare the flow capture ratios at all velocities and for all structures, we normalized the flow capture ratio at a given velocity by the maximum flow capture ratio obtained for all structures at the same velocity (figure 8b). The justification for taking a different normalizing flow capture ratio at each velocity is that we want to compare all the structures at a fixed velocity to determine which geometry is the more efficient at this given velocity. Fitting an overarching sigmoidal model to the observed trends enables us to explore a wider range of structural architectures. We can see that there is no design that is optimized for all velocities. For designs with many rami, the flow capture ratio is very low at low velocities whereas designs with few rami perform poorly at high velocities. Actually, each design has a maximum flow capture ratio spanning a range of air velocities. The maximum flow capture ratio of the natural design (50 rami) is centred on the moth flight velocities.

The architecture of real antennae has some slight differences from the surrogate ones which might be important. Indeed, rami of a pectinate antenna are not completely parallel, as can be seen in figure 1a, and this seems to be the case for several species of moths [42]. The rami are almost equidistant at their base but get closer towards their tips, where they overlap. This change in inter-rami distance might have consequences for the flow capture ratio. At the base, the design is similar to our structure with 50 rami and the flow capture ratio is thus correctly determined. However, as the rami overlap at the tip,

we end up with a structure that resembles that of 25 rami. In between, we can assume a continuum between the two extreme configurations. As a result, we can assume that the change in distance between the rami extends the range of air velocities towards the low velocities, where the antenna is efficient (figure 8b). We, therefore, conclude that the overall architecture of pectinate antennae of insects is a robust design well suited to function over a wide range of flow velocities.

Data accessibility. The templates for printing the antennae are at <https://doi.org/10.6084/m9.figshare.12287963.v1>.

Authors' contributions. M.J.-B. ran the experiments, analysed the data and wrote the first draft of the article. T.S. provided technical support for running the PIV set-up and post-processing the data. He was also involved in analysing the data and contributed to the writing of the article. G.K. and J.C. co-conceived the design of the study, contributed to the data analysis and co-authored the article.

Competing interests. We declare we have no competing interest.

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